

CompositionMeasure for RenyiDivergence

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This proof resides in “**contrib**” because it has not completed the vetting process.

Proves soundness of the implementation of **CompositionMeasure** for **RenyiDivergence** in **mod.rs** at **commit f5bb719** (outdated¹).

1 Hoare Triple

Precondition

Compiler-verified

Types matching pseudocode.

Caller-verified

None

Pseudocode

```
1 class CompositionMeasure(RenyiDivergence):
2     def composability( #
3         self, adaptivity: Adaptivity
4     ) -> Composability:
5         return Composability.Concurrent
6
7     def compose(self, d_mids: Vec[tuple[Self_Distance, u32]]) -> Self_Distance:
8         # merge equal curves so that each is evaluated once, not once per copy
9         groups = OrderedCounts() #
10        for d_mid, k_i in d_mids:
11            groups.add(d_mid, k_i)
12
13        def curve(alpha: float) -> float: #
14            epsilons = [
15                d_mid(alpha).inf_mul(f64.from_(k)) #
16                for d_mid, k in groups
17            ]
18
19            d_out = 0.0
20            for d_mid in epsilons:
21                d_out = d_out.inf_add(d_mid)
22            return d_out
23
24        return Function.new_fallible(curve)
```

¹See new changes with `git diff f5bb719..d7b4364b rust/src/combinators/sequential_composition/mod.rs`

Postcondition

Theorem 1.1. `composability` returns `Ok(out)` if the composition of a vector of privacy parameters `d_mids`, where each `d_mid` is charged `k` times, is bounded above by `self.compose(d_mids)` under adaptivity, adaptivity and out-composability. Otherwise returns an error.

Proof. The new curve constructed on line 13 composes all epsilon parameters at a given fixed `alpha`. By the postconditions of `InfAdd` and `InfMul` we have that for any choice of `alpha`, $\sum_i k_i d_mid_i(\alpha) \leq \text{compose}(d_mids)(\alpha)$.

Adaptivity	Sequential	Concurrent
Non-Adaptive	Proposition 1[Mir17]	Theorem 2[Lyu22]
Adaptive	-	-
Fully-Adaptive	Theorem 4.3[FZ22]	Theorem 1.22[VW21]

This table is reflected in the implementation of `composability` on line 2.

□

References

- [FZ22] Vitaly Feldman and Tijana Zrnic. Individual privacy accounting via a renyi filter, 2022.
- [Lyu22] Xin Lyu. Composition theorems for interactive differential privacy, 2022.
- [Mir17] Ilya Mironov. Rényi differential privacy. In *2017 IEEE 30th Computer Security Foundations Symposium (CSF)*, page 263–275. IEEE, August 2017.
- [VW21] Salil Vadhan and Tianhao Wang. Concurrent composition of differential privacy, 2021.